

1. Fried food

S1: Eating fried food frequently leads to accumulation of bad cholesterol in the body leading to heart diseases.

S2: Ron consumes fried food every week.

Which of the following needs to be implied for the given conclusion to be valid?

1. Fried food is the only way to increase bad cholesterol in the body.
2. Consuming fried food once every week is considered frequent consumption of fried food.
3. Ron is not concerned about his heart health.

Conclusion: Ron suffers from a heart disease.

Answer Options

Select any one option

[Clear Answer](#)

☐ Only 1

☒ Only 2

☐ Only 3

☐ Both 1 and 2

2. Drip irrigation for lawns

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Next >

Gardener: When adding lawns for the new house, we should install drip irrigation system in addition to the sprinklers. This will ensure that the lawns don't dry out in case someone forgets to water the lawn and reduce damage in the garden. The cost of installing drip irrigation system is relatively lower when compared to redoing the lawns due to the damage caused by under watering.

House owner: Installing drip irrigation systems are very expensive and unnecessary and seem like a wasteful expense.

Which of these choices, if true, could best be used to support the Gardener's proposal in the face of the House owner's objection?

Answer Options

Select any one option

Clear Answer

☐ The use of sprinklers can still reduce the longevity of the lawns due to under watering.

☐ Severe rains can cause the lawns to rot

☐ During a snow storms, all the lawn gets slightly damaged.

☒ Drip irrigation system can save the lawns when nobody is there to water it. ✓

3. A circular plot

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A circular plot has been allotted by the government for the construction of an embassy building. The government decides to increase the area of the plot by 21%.

What would the percentage increase in the circumference of the plot be as a result of this?

Answer Options

Select any one option

[Clear Answer](#)

☒ 10%

☐ 15%

☐ 20%

☐ 11%

Aptitude ends in 29 min

Aptitude

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

Software

Starts in 29 min

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4. Profits made by Mrs. Lindsay

< Previous

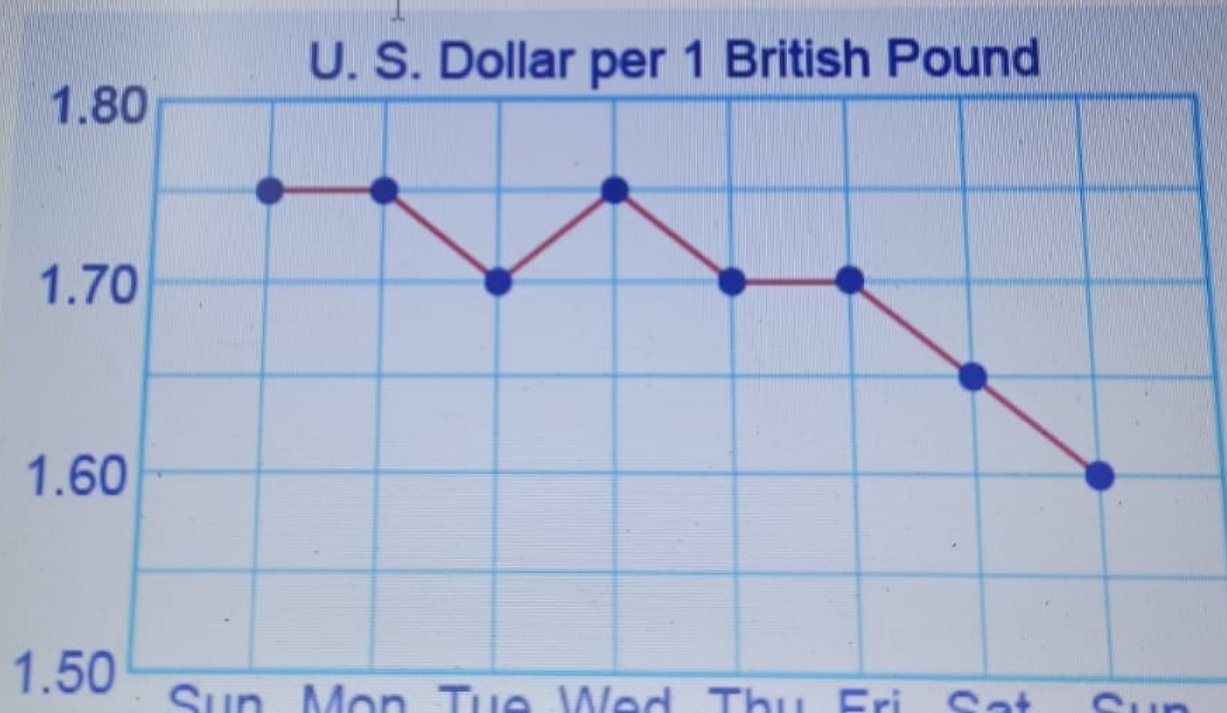
Next >

The given line graphs show the exchange rate dynamics during one week.

On Tuesday, Mrs. Lindsay bought 200 British Pounds using US Dollars. On Wednesday, she exchanged these 200 British Pounds back into US Dollars. What profit did Mrs. Lindsay make?

Note: Assume that there was zero transaction cost.

10



The two grids A and B given below follow a certain pattern. Analyse A and determine the correct value of '?' in B.

3	9	4
8	2	14
3	12	6

A

4	13	5
?	4	15
2	12	6

B

Answer Options

Select any one option

[Clear Answer](#)

☐ 10

☐ 11

6. Cats and dogs

Given below are 4 statements followed by 4 conclusions numbered 1 to 4. You have to take the given statements to be true even if they seem to be at variance from commonly known facts. Choose the most appropriate option for the given conclusions that logically follows from the given statements, disregarding commonly known facts.

Statements:

Some cats are dogs. No dog is a frog. All frogs are bats. Some bats are eagles.

Conclusions:

1. Some frogs are not eagles.
2. No dog is a bat.
3. Some cats are bats.
4. No eagle is a cat.

Answer Options

Select any one option

- ☐ Both conclusion 2 & 4 follow
- ☐ Conclusions 2, 3 & 4 follow
- ☐ Only conclusion 1 follows
- ☐ No conclusion follows

Software

Starts in
00:14:40

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Software](#)



[Submit
test](#)

16. A leading research institute

XYZ is a leading research institute based out of India that is working on making vaccinations for COVID-19. The institute has recently been losing a lot of its bids, and this has resulted in a critical loss of revenue.

The board of directors suspect industrial espionage and decide to investigate the Lead Scientist, Lead Engineer, and the VP. It is brought to light that the details of a certain bid in the summer of the previous year were leaked to their arch-rivals Y by the mole in the institute. Thus the board questions the Lead Scientist, Lead Engineer, and the VP and asks them to provide their alibis for the date on which the contract bid was made. The respective people give the following alibis.

Analyze the same and identify the mole.

Lead Scientist: I was in Mumbai to spend the summer with my family.

Lead Engineer: I had gone to Sydney to enjoy the summer with my wife.

VP: I had gone to Bengaluru to perform market research for expanding the company.

Answer Options

Select any one option

Clear Ans

☐ Lead Scientist

17. Selling Ras Malai

The given table shows the ratio of the marked price and cost price of the items and the discount % and the profit earned by Martin.

It is given that the cost price of Ras malai is as same as the cost price of Rasgulla. The marked price of Rasgulla is 25% more than the cost price.

Find the profit earned by Martin after selling the Ras Malai if the marked price of Ras Malai is 56.25% more than the cost price of Ras malai.

Items	Marked Price: Cost Price	Discount %	Profit (in Rs)
Gulab Jamun	25:18	-	63
Rasgulla	-	8%	48
Ras Malai	-	20%	-
Kaaju Katli	6:5	-	-
Laddu	25:18	-	45

80

Answer Options

Select any one option

Clear

Aptitude ends in 26 min

Aptitude

- 1 2 3
4 5 6
7 8 9
11 12
14 15
17 18
20

26 min

18. Dogs and cats

It is given that a dog barks once every two minutes and a cat meows on average once every five minutes. The number of times a dog would bark in 24 hours is approximately equal to what percent greater than the number of times a cat would meow in that same period?

Answer Options

Select any one option

Clear Answer

☒ 150%



☐ 250%

☐ 100%

☐ 50%

19. Analyse the two grids

< Previous

Next >

The two grids A and B given below follow a certain pattern. Analyse A and determine the correct value of '?' in B.

2	11	7
5	13	6
8	13	3

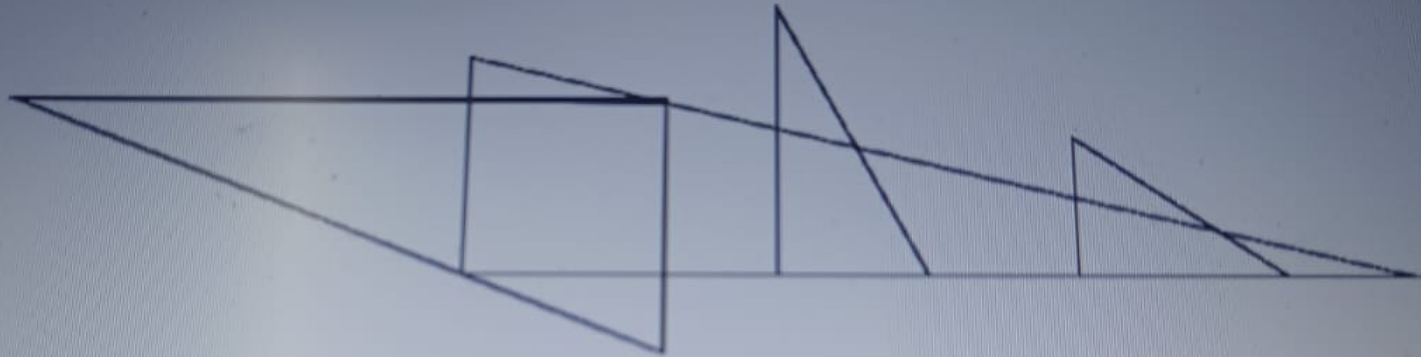
A

7	17	8
10	?	4
5	12	5

B

20. Find the triangles present in the image

How many triangles are present in the given image?



14

Answer Options

Select any one option

☐ 10

☐ 11

Drip irrigation for lawns

Gardener: When adding lawns for the new house, we should install drip irrigation system in addition to the sprinklers. This will ensure that the lawns don't dry out in case someone forgets to water the lawn and reduce damage in the garden. The cost of installing drip irrigation system is relatively lower when compared to redoing the lawns due to the damage caused by under watering.

House owner: Installing drip irrigation systems are very expensive and unnecessary and seem like a wasteful expense.

Which of these choices, if true, could best be used to support the Gardener's proposal in the face of the House owner's objection?

Answer Options

Select any one option

- ☐ The use of sprinklers can still reduce the longevity of the lawns due to rot
- ☐ Severe rains can cause the lawns to rot
- ☐ During a snow storms, all the lawn gets slightly damaged.
- ☒ Drip irrigation system can save the lawns when nobody is there to water it.

Software

Starts in
00:14:20

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Software](#)

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test](#)

Laddu	25:18
-------	-------

Answer Options

Select any one option

☐ Rs. 60

☒ Rs. 80 ✓

☐ Rs. 100

☐ Rs. 120

A and B
Two grids A and B given below follow a certain pattern. Analyse A and determine the correct value of '?' in B.

3	9	4
8	2	14
3	12	6

A

4	13	5
?	4	15
2	12	6

B

Answer Options

Select any one option

☐ 10

☐ 11

☐ 12

 Submit test



software
Starts in
00:13:28
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15
18
20

ne profession of E

There are six people A, B, C, D, E, and F in a family. They are an Engineer, Manager, Painter, Jeweler, Doctor, and Lawyer by profession.

1. The doctor is the grandfather of F, who is an Engineer.
2. The manager D is married to A.
3. C, the jeweler, is married to the Painter.
4. B is the mother of E and F.
5. E is a bachelor.

Using the given information, determine the profession of E?

Note: It is given that no person other than A, B, C, D, E, and F is involved in the relationships described above.

Answer Options

Select any one option

☐ Engineer

☐ Jeweler

☒ Manager

☐ Lawyer

Software

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00:14:00

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test

Row of athletes

In a row of 30 athletes facing North, A1 is 13th from the left end and 6th to the right of A2. What is the position of **A2** from the **right end** of the row?

Answer Options

Select any one option

☒ 24th

☐ 7th

☐ 15th

☐ 20th

Software

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00:14:56

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Software](#)



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test](#)

Selling Ras Malai

The given table shows the ratio of the marked price and cost price of the items and the profit earned by Martin.

It is given that the cost price of Rasmalai is as same as the cost price of Rasgulla. The marked price of Rasgulla is 25% more than the cost price.

Find the profit earned by Martin after selling the Ras Malai if the marked price of Ras Malai is 56.25% more than the cost price of Rasmalai.

Items	Marked Price	Cost Price	Discount %	Profit (in Rs)
Gulab Jamun	25.18	-	-	63
Rasgulla	-	6.5	8%	48
Ras Malai	-	25.18	20%	-
Kaju Katli	-	-	-	-
Laddu	-	-	-	45

Answer Options

Select any one option

- ☒ Rs. 60
- ☐ Rs. 80
- ☐ Rs. 100
- ☐ Rs. 120

Software

Starts in
00:11:27

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Software](#)

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Aptitude

- 1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Software

Starts in 15 min

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Software >

A rectangular plot has been allotted by the government for the construction of a embassy building. The length of the plot is increased by 10% to equip a cafeteria.

By what percent should the breadth of the plot be decreased such that the area of the plot is same as before?

$$20y = 1.1x \times py$$

Answer Options

Select any one option

☐ 120/13

☒ 100/11 ✓

☐ 120/11

☐ 100/9

10. A mocktail

Bob is preparing a mocktail for his friends by mixing 10 parts water and 5 parts of a mocktail mix in a large jug. How much of the mixture should he draw out of the jug and replace it with the mocktail mix to ensure that the mixture is half water and half mocktail mix?

Note: Assume that the jar contains 15 litres of the mixture.

Answer Options

Select any one option

☒ 1/4

☐ 1/8

☐ 1/2

☐ 1/3

Software

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00:13:12

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Software](#)

 Submit
test

Disappointed customers

Jake is a butcher. Given below is his selling procedure:

1. Get the order from the customer
2. Weigh the meat on the goods pan against the weights put on the weighing pan
3. Give it to the customer.

However, certain criteria need to be met.

1. The order can be anything in kilograms as long as it is a whole number.
2. The weight cannot be placed in the goods pan in the weighing machine.
3. Jake has to weigh the meat at one go and can't do it in parts.
4. If Jake can't measure the weight, he sends them back.

Jake has weights of 1, 2, 4, 8, 16, 32, 64 kilograms. Customers start coming to Jake's shop right from the morning. A customer buys the same weight of meat as his number. For instance, customer number 32 buys 32 kilograms of meat. A total of 128 customers come to Jake's shop on Wednesday.

How many customers went away disappointed that day?

Answer Options

Select any one option

- ☐ 0
- ☒ 1
- ☐ 2
- ☐ 3

Software

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00:10:23

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Software](#)



Submit
test

7. Ceaser's money

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A business man named Ceaser has lent 10 million dollars at an interest rate of 40% per annum to his son for his new business.

After how many years would the money lent by Ceaser double itself if it is compounded semi-annually?

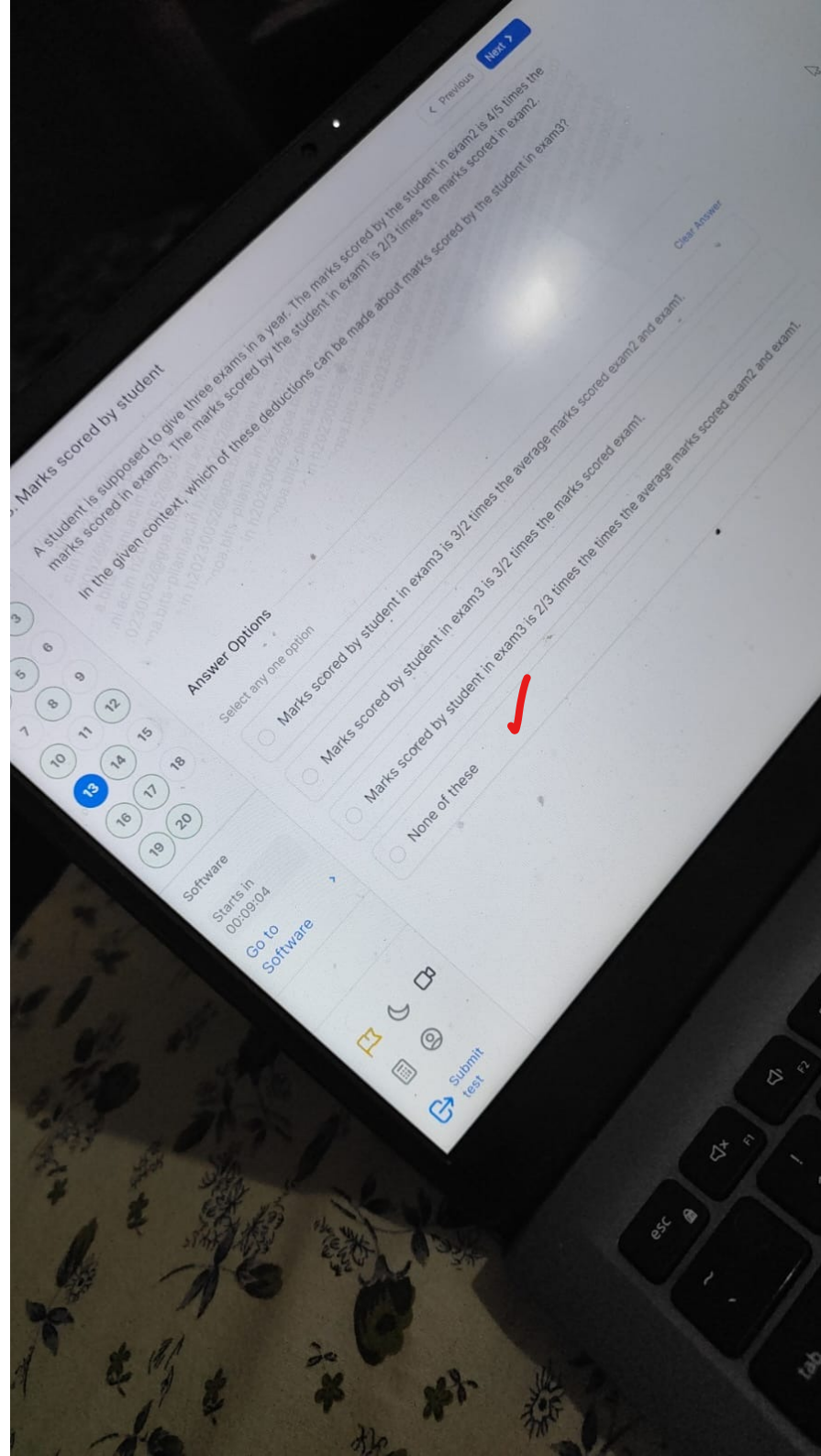
less than 2 y

Answer Options

Select any one option

- ☐ In exactly 2 years
- ☐ In more than 2 years

[Clear A](#)



6. Executing a function

Predict the output of the following code snippet.

```
void encrypt (char *plainText, char *cryptText) {  
    int i;  
    for (i=0; plainText[i] != 0; i++) {  
        cryptText[i] = (plainText[i] <= 'm')? (plainText[i] + 13) : (plainText[i] - 13);  
    }  
    cryptText[i] = 0;  
}  
  
main() {  
    char t[10], s[10];  
    encrypt("hello",t);  
    encrypt(t,s);  
    printf("%s %s", s, t);  
}
```

Answer Options

Select any one option

- ☐ Run-time error
- ☐ Compilation error due to incorrect syntax
- ☐ uryyb hello
- ☒ hello uryyb ✓

3. The UART communication protocol

You are using the following 8051 code to send a char using the UART communication protocol.

```
void UART_TxChar(char ch)
{
    SBUF = ch;
    while(TI==0);
}
```

What is missing in the code?

Answer Options

Select any one option

- ☒ The TI flag is not cleared ✓

4. The time complexity

What is the time complexity of the piece of

```
int p = 0;  
for (i = 0; i < n; i++) {  
    for (j = n; j > i; j--) {  
        p = p + i + j;  
    }  
}
```

Answer Options

Select any one option

☐ $O(n)$

☒ $O(n*n)$ ✓

☐ $O(n \log n)$

☐ $O(\log n)$

are ends in 44
min

1. Executing a set of micro instructions

Which of these address instructions does the following set of micro instructions execute?

LOAD A
ADD B
STORE T
LOAD C
ADD D
MUL T
STORE X

$$(A+B) \times (C+D)$$

Answer Options

Select any one option

☒ $Y = (A+B) * (C+D)$

5. Rotating elements

You are given an array of size N and asked to rotate it by d , denoting the number of rotations.

What is the time complexity of performing this operation by rotating the elements one by one?

if $d \leq N$ ✓
else N

Answer Options

Select any one option

☒ $O(N)$

2. The printf statement

Predict the output returned by the code snippet.

```
#include <stdio.h>
int main()
{
    int var=200;
    {
        int var=300;
        printf("%d...",var);
    }
    printf("%d",var);
    return 0;
}
```

Answer Options

Select any one option

☐ 200...200

☐ 100...100

☒ 300...200

Software ends in 42 min

8. A LRU algorithm

Assume that N is assigned to the number of pages in the memory and which is the size of the LRU algorithm.

What can be said about the page faults that occur in the LRU algorithm with respect to the Optimal (OPT) algorithm?

Answer Options

Select any one option

[Clear Answer](#)

- ☒ The LRU can never result in more than N -times more page faults than Optimal (OPT) algorithm ✓
- ☐ The LRU always results in more than N -times more page faults than Optimal (OPT) algorithm
- ☐ The LRU always results in N -times less page faults than Optimal (OPT) algorithm
- ☐ The LRU always results in less page faults equal to N -time page faults in Optimal (OPT) algorithm


```
while (!stk.isEmpty())  
{  
    rev = rev + stk.pop();  
}  
return rev;  
}
```

9

```
public String reverse(String input)  
{  
    for (int i = 0; i < input.length(); i++)  
    {  
        stk.push(input.charAt(i));  
    }  
    String rev = "";  
    while (!stk.isEmpty())  
    {  
        rev = rev + stk.pop();  
    }  
    return rev;  
}
```



```
public String reverse(String input)  
{  
    for (int i = 0; i < input.length(); i++)  
    {  
        stk.push(input.charAt(i));  
    }  
    String rev = "";  
    while (!stk.isEmpty())  
    {  
        rev = rev + stk.pop();  
    }  
}
```

it test

```
public String reverse(String input)  
{  
    for (int i = 0; i < input.length(); i++)  
    {  
        stk.push(input.charAt(i));  
    }
```

```
public String reverse(String input)
{
    for (int i = 0; i < input.length(); i++)
    {
        stk.push(input.charAt(i));
    }
    String rev = "";
    while (!stk.isEmpty())
    {
        rev = rev + stk.pop();
    }
}
```

```
public String reverse(String input)
{
    for (int i = 0; i < input.length(); i++)
    {
        stk.push(input.charAt(i));
    }
    String rev = "";
    while (!stk.isEmpty())
    {
        rev = rev + stk.pop();
        stk.pop();
    }
    return rev;
}
```


9. Reversing a word

You want to reverse a word using stacks.

Which of these code snippets can you use to successfully achieve this?

Answer Options

Select any one option

☒

```
public String reverse(String input)
{
    for (int i = 0; i < input.length(); i++)
    {
        stk.push(input.charAt(i));
    }
    String rev = "";
    while (!stk.isEmpty())
    {
        rev = rev + stk.peek();
    }
    return rev;
}
```

☐

```
public String reverse(String input)
{
    for (int i = 0; i < input.length(); i++)
    {
        stk.push(input.charAt(i));
    }
    String rev = "";
    while (!stk.isEmpty())
    {
        rev = rev + stk.pop();
    }
}
```

10. A virtual memory system

An OS has a virtual memory system that implements the FIFO page replacement algorithm. Assume that for an arbitrary page access pattern, the number of page frames in the main memory is increased.

What will happen as a result of doing this?

Answer Options

Select any one option

Clear Ans

- ☐ The number of page faults will increase
- ☐ The number of page faults will decrease
- ☒ The number of page faults might increase sometimes

LRU algorithm

Assume that N is proportional to the number of pages in the managed pool when working with a LRU algorithm.

What can be said about the page faults that occur in the LRU algorithm with respect to the Optimal (OPT) algorithm?

Answer Options

Select any one option

- ☐ The LRU can never result in more than N -times more page faults than Optimal (OPT) algorithm
- ☐ The LRU always results in more than N -times more page faults than Optimal (OPT) algorithm
- ☐ The LRU always results in N -times less page faults than Optimal (OPT) algorithm
- ☐ The LRU always results in less page faults equal to N -time page faults in Optimal (OPT) algorithm

Clear Answer



Submit test

1. Executing a set of micro instructions

Which of these address instructions does the following set of micro instructions execute?

LOAD A
ADD B
STORE T
LOAD C
ADD D
MUL T
STORE X

Answer Options

Select any one option

- ☐ $X = (A+B) * (C+D)$ ✓
- ☐ $X = (A+B) * C+D$
- ☐ $X = A+(B * C)+D$
- ☐ None of these



Submit
test

5. The UART communication protocol

You are using the following 8051 code to send a char using the UART communication protocol.

```
void UART_TxChar(char ch)
{
    SBUF = ch;
    while(TI!=0);
}
```

What is missing in the code?

Answer Options

Select any one option

- ☒ The TI flag is not cleared
- ☐ The TI flag is initialised
- ☐ Both Choice 1 and 2
- ☐ Neither Choice 1 nor 2

Submit
test

7. Executing a function

Predict the output of the following code snippet.

```
void encrypt(char *plainText, char *cryptText) {  
    int i;  
    for (i=0; plainText[i] != 0; i++) {  
        cryptText[i] = (plainText[i] <= 'm') ? (plainText[i] + 13) : (plainText[i] - 13);  
    }  
    cryptText[i] = 0;  
}
```

```
main() {  
    char t[10], s[10];  
    encrypt("hello", t);  
    encrypt(t, s);  
    printf("%s %s", s, t);  
}
```

Answer Options

Select any one option

☐ Run-time error

☐ Compilation error due to incorrect syntax

☐ uryyb hello

☒ hello uryyb



4. The time complexity

What is the time complexity of the pic

```
int p = 0;
for (i = 0; i < n; i++) {
    for (j = n; j > i; j--) {
        p = p + i + j;
    }
}
```

3

6

9

Answer Options

Select any one option

☐ $O(n)$ ☐ $O(n*n)$ ☐ $O(n \log n)$ ☐ $O(\log n)$